Innovation & Regulation

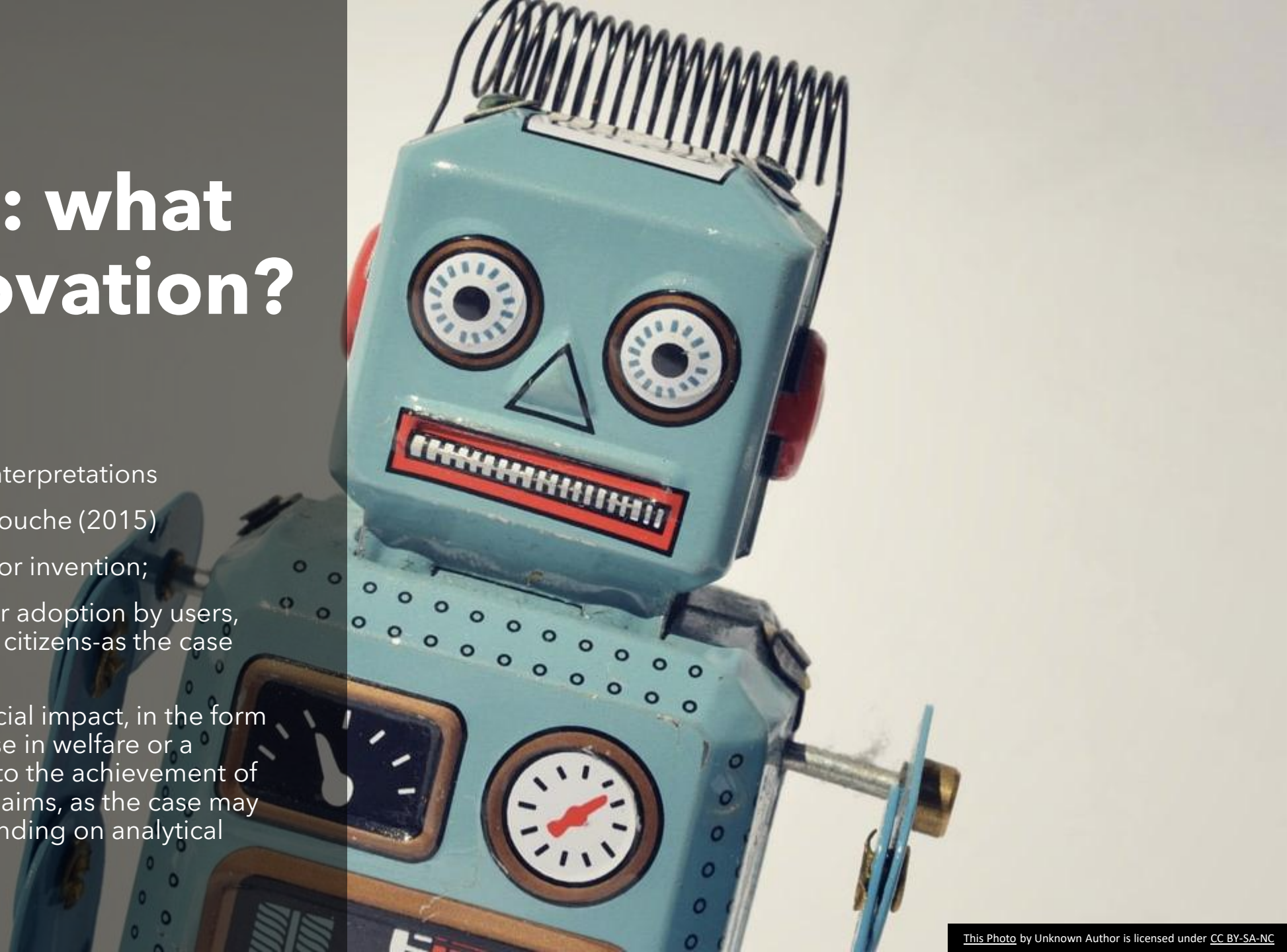
Introduction (Part 2)

Dr. M. K. (Konrad) Borowicz

Recap: what is innovation?

- Many different interpretations
- Butenko and Larouche (2015)

1) A novel idea or invention;
2) Its diffusion or adoption by users, customers or citizens-as the case may be;
3) A positive social impact, in the form of the increase in welfare or a contribution to the achievement of public policy aims, as the case may be and depending on analytical perspective.



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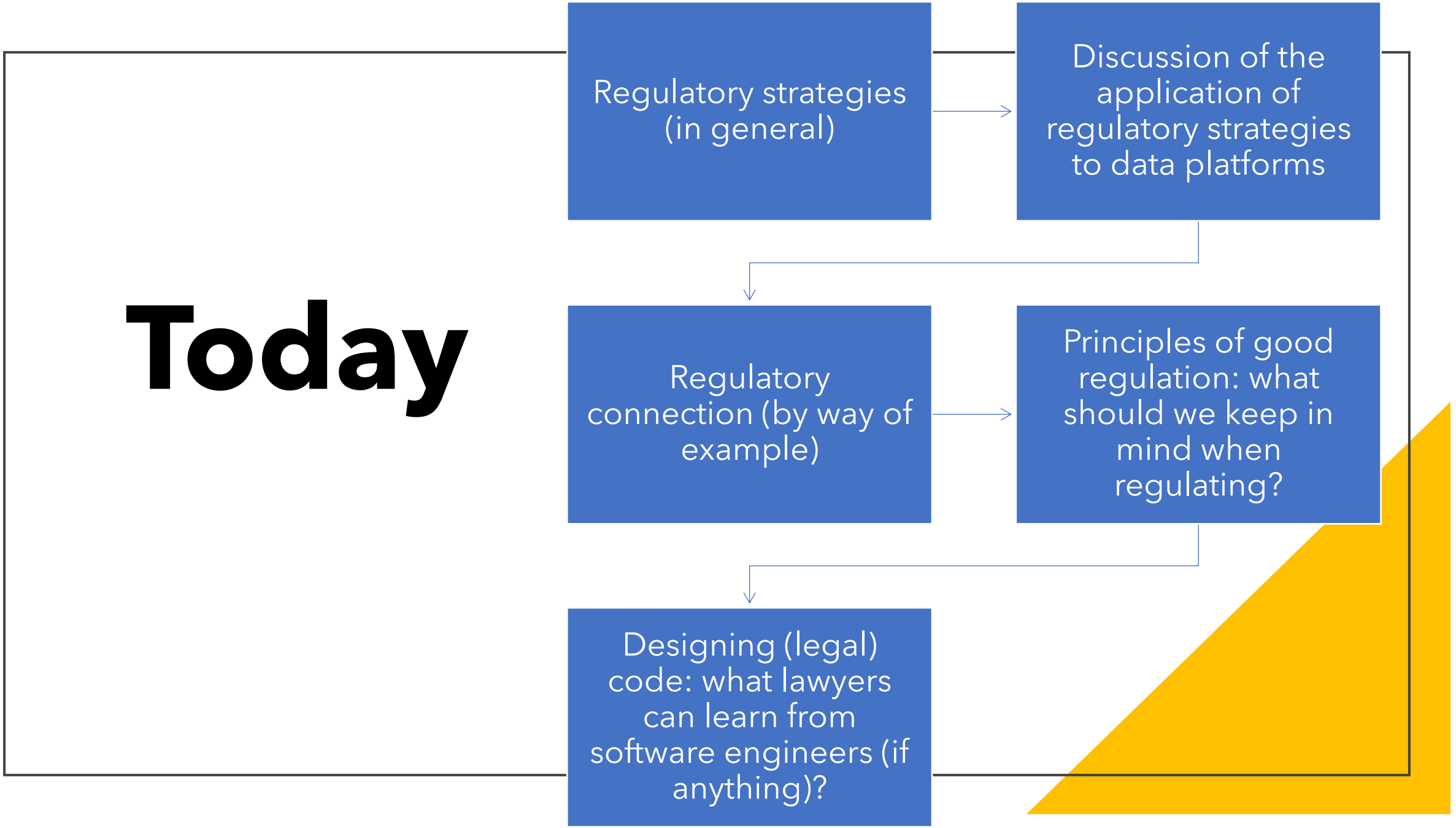
Recap: what is REGULATION?

"any instrument (legal or non-legal, governmental or non-governmental) … that is designed to channel behavior" (Brownsword and Somsen, 2009);

"intentional influencing of someone's or somethings behavior" (Black 2005)

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Today


Regulatory strategies (in general)
Discussion of the application of regulatory strategies to data platforms
Regulatory connection (by way of example)
Principles of good regulation: what should we keep in mind when regulating?
Designing (legal) code: what lawyers can learn from software engineers (if anything)?

Regulatory strategies

- There are plenty of regulatory strategies with strengths and weaknesses
- Governments usually apply a combination of different instruments;
- We discuss a few examples based on Baldwin, Cave & Lodge.



Command & control

- Imposing standards backed by (criminal) sanctions (e.g. Clean Air Act Sect. 7411 and 7413);
- Pros:
  - fixed standards,
  - unacceptable behavior is prohibited,
  - seen as **protective of the public('s rights**).
- Cons:
  - prone to **regulatory capture** (discretion),
  - difficulties to design targeted rules,
  - rules may become outdated,
  - intervenes in management,
  - information requirements,
  - enforcement difficulties.





Incentive-based

- Regulating by economic incentives (e.g. tax reductions subsidies but also penalties);
- Pros:
  - less discretion for regulators, more flexibility for regulated parties,
  - lower information requirements and
  - low cost of enforement;
- Cons:
  - difficulty to predict effects,
  - incentives operate mechanically,
  - socially harmful behavior not condemned.

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Market-harnessing controls

Channelling market forces to influence competition to ensure market provides adequate services

- Competition

Pros:

- apply across sectors,
- lower entry barriers for firms than in C&C,
- lower level of intrusion in fims' internal decisions,
- lower burden on public enforcement system.

Cons:

- more uncertainty (no clear detailed ex ante rules),
- courts are slow to generate guidance,
- enforcement ex post and too late.

Other examples

- Tradable permits

Disclosure

- Structuring the disclosure of **information** (making sure it is available);
- Pros:
  - not heavily interventionist (deference to consumer preferences);
- Cons:
  - consumers may make mistakes,
  - Information is costly (to produce and consume),
  - 'irrational behavior',
  - inaccurate information,
  - too great risks to leave them unregulated.



Rights and liablities

- Allocation of property rights will induce socially desirable behavior
- If the violation of such rights is associated with some form of liablity …
- …people will be disincentivized to violate such rights because of the potential liablity to pay damanges when sued by holders of rights
- Problems:
  - not always easy to allocate property rights
    - public goods ⟩ tragedy of the commons
    - private goods ⟩ tragedy of the anti-commons
  - it can be efficient to engage in socially *un*desirable behavior if damages not set right;
  - insurance can decrease deterrence effect;
  - enforcement costs can be high.

Risk-based

- Prioritizing of regulatory actions in accordance with an **assessment of the risks** an activity will present to the regulatory body's achieving of its objectives
- Requires development of **risk-scoring mechanisms**
- Very popular in the last two decades
- Pros:
  - Helps avoid overregulation
  - Increases 'regulatory connection'
- Cons:
  - Not a neutral exercise - regulator's risk tolerance is often driven by political considerations
  - Some acivities/firms may 'fly under the radar'
  - Requires significant degree of cooperation from the industry

Design solutions

- **Limit opportunities** for undesirable behaivor
- In physical environment but also the (legal) market **design**
- Also, technology as regulation (e.g. by asking software designers to build-in certain elements into software code in pursuit of public regulatory objectives)
- **Nugding** as a policy tool
  - structuring the architecture of decisions ('choice architectures'), so that it is easier for consumers or others to act in ways that are beneficial (e.g. put healthy food in most visible spots in canteens).
- Pros:
  - decision makers are free to behave,
  - less costly,
  - combination element of paternalism with freedom of choice;
- Cons:
  - ethical concerns, who decides what is good or best for consumer,
  - threat of manipulation,
  - low accountability and transparency.



Mock exam question: YouTube content moderation

- Pick a regulatory strategy and consider
  - What are the key features and assumptions of this regulatory strategy?
  - How might this strategy be applied to regulate content on YouTube, and what are its strengths and limitations?
  - What are some potential unintended consequences or trade-offs of this strategy, and how can they be addressed?

Gonzalez v. Google (2023)

- Nohemi Gonzalez was a 23-year-old American student killed in the 2015 ISIS attack in Paris.
- The family argued that Google should be held liable for its platform, YouTube, recommending ISIS videos to its interested users (aiding and abetting terrorists)
- Google relies on Section 230 of the Communications Decency Act, passed by Congress in 1996 to escape liability

Section 230 CDA

Regulatory disconnect?

- “The Twenty-Six Words That Created the Internet”
- "No provider or user of an interactive computer service shall be treated as the publisher or speaker of any information provided by another information content provider." (47 U.S.C. § 230(c)(1)).
- Assumption that platforms simply *transport* the work of a third-party
- But this assumption reflect how the companies have utilized advances in digital technology?
- Internet of 1996 and 2023 is quite different
- Algorithms…

Regulatory disconnect

- The ‘pacing problem’
- Brownsword: challenge of **regulatory connection** refers to the widening gap between the current regulatory environment based upon the ‘technological landscape of the past’ and the occurring innovations revolutionizing that landscape.
- Several possible variants of ‘regulatory disconnect’
  - Regulatory voids or gaps (extreme case, black swan technologies)
  - Ambiguity in the application of existing regulations
  - Overregulation or underregulation
  - Regulatory obsolecence

Horizontal dimension of (dis)connection

- At what stage of the innovation should the regulation connect with it
  - If you connect too early, you can stifle it
  - If too late, it can run away…
- Preferred approach:
  - risk-based regulation
  - precautionary principle
- Experimental approach to regulation/governance
  - Regulatory sandboxes

Vertical dimension of (dis)connection

- Technology specific
  - Electric cars
  - AI
- Technology neutral
  - property rights
  - competition
  - consumer law
- What form/level?

Institutional dimension of (dis)connection

- Importance of regulatory authorities
  - Independent from market actors
  - Accountable
  - Holds a well-defined legislative mandate
  - Has the resources
  - Required to respect certain (constitutional) principles
- Well-positioned to address
  - the vertical dimension of (dis)connection by adopting a layered approach to regulation – from general principles (technology neutral) to more specific rules
  - the horizontal dimension of (dis)connection by adopting general principles, which do not need to be changed frequently and can guide the development of technology

Self-regulation, ethics?

- Back to Section 230 DCA
- US Supreme Court did not to change the law
- If no liability, then what?
  - Corporate self-regulation and proactive application of **ethics** to emerging technology is a must?
  - YouTube has large content moderation teams…
- Role of the contractual relationship between provider and users
- We alreadys see contractual attempts at concretising responsibility and rules of engagement but
  - Information asymmetries
  - Unequal bargaining power

EU's Digital Services Act

- In 2000, the European Union adopted the Electronic Commerce Directive.
- Like Section 230, the eDirective protected online platforms from liability for the passive retransmission of third-party content.
- In 2022, the EU's Digital Services Act (DSA), while leaving the eDirective undisturbed, established a "duty of care" for online platforms, with the most expansive duties reserved for the largest platforms.

EU's Digital Services Act

- At the heart of the DSA are **disclosure** requirements re
  - algorithmic and
  - human content moderation.
- **Disclosure** re operation of recommendation algorithms
  - standards re recommendation (not use individual profiling as its basis).
- For large platforms, also **risk-based** where assessment of risks "stemming from the design, functioning and use of their services" delegated to platforms who are also under the obligation to "deploy the necessary means to diligently mitigate the systemic risks identified."

Mock exam question: Google search

- Pick a regulatory strategy and consider
  - What are the key features and assumptions of this regulatory strategy?
  - How might this strategy be applied to regulate search on Google, and what are its strengths and limitations?
  - What are some potential unintended consequences or trade-offs of this strategy, and how can they be addressed?

EU Digital Markets Act

- The EU Digital Markets Act (DMA, 2022) imposes ex-ante obligations on very large "gatekeeper" search engines to share user query and click data with other search engines.
- DMA Art 6(11) states that gatekeepers "shall provide to any third-party providing online search engines, at its request, with access on fair, reasonable and non-discriminatory terms to ranking, query, click and view data in relation to free and paid search generated by end users on its online search engines".
- **Market-harnessing** approach with **C&C** characteristic
- Instead of a complicated lawsuit against Google, the Commission simply commanded a solution that made sense in view of the data-driven network effects:
  - more users improve the quality of search and therefore attract even more users to the search engine; and
  - more users generate more data that improve the quality of the service.

AI Act

- EU's AI Act will impose rules on AI systems depending on the **risks they present** (as envisaged by the provider)
  - High risk (all systems intended to be used in one of the areas specified in annex III)
  - No risk (or, in the case of deep fakes and chatbots: a low-risk category with **disclosure** as the ultimate regulatory answer).

AI Act

- But for general purpose AI, it is not the provider but rather the user who determines how they will use the system.
- Too late from the point of view of society and fundamental rights?
  - "Unless safety is considered from the outset, any potential biases, privacy violations, unlawful uses of content or other instances of unfairness in the data or the model will trickle down into a myriad of possible future applications."

Risk-based the way forward?

- But can we anticipate what the risks are ex ante?
- Systemic risk monitoring?
- Compare with the DMA requirements

What is ‘good’ regulation?

- When discussing this question people generally focus a combination of the following considerations:
  - Is the action or regime supported by legislative authority?
  - Is there an appropriate scheme of accountability?
  - Are procedures fair, accessible, and open?
  - Is the regulator acting with sufficient expertise?
  - Is the action or regime efficient?
- Measuring regulatory quality
  - Administrative burden, regulatory complexity (European approach)
  - Outcomes and net benefits for citizens (US, Canada)

Case study in regulatory complexity

- Li et al (2014), *Law is Code: A Software Engineering Approach to Analyzing the US Code*
- Commentators cite numerous problems with the U.S. Code, including:
    1. many of the new laws passed since 1926 are often shoehorned awkwardly into pre-existing titles;
    2. Congress often pays little or no attention to existing laws when enacting new legislation, which makes it difficult for U.S. Code editors to keep statutes that relate to similar subjects together; and
    3. the increasingly voluminous body of legislation since 1926 has produced many obscure, obsolete, and redundant provisions, archaic and inconsistent language, and statutory errors

Software-engenering approach?

- Some analogies between sofware and legal code.
- Both represent **rules or instructions** for, respectively, machines and humans
- Both software engeeners and lawyers are concerned with the code, respectively, software and legal, be **readable and easy to understand**
- These requirements suggest that “the tools used by software engineers to track progress, monitor potential vulnerabilities, or simply gain an understanding of an existing software codebase may be useful for serving the same functions when applied to legal code.”

Software-engenering approach?

- Four categories of issues that affect the understandability of the legal code:
  - conciseness (LOC?),
  - change (function retained),
  - coupling (object-oriented programming, modularity), and
  - complexity (conditional statements).
- Software engineers and lawyers have a lot in common! Maybe?

Article 34

Risk assessment

1. Providers of very large online platforms and of very large online search engines shall diligently identify, analyse and assess any systemic risks in the Union stemming from the design or functioning of their service and its related systems, including algorithmic systems, or from the use made of their services.

They shall carry out the risk assessments by the date of application referred to in Article 33(6), second subparagraph, and at least once every year thereafter, and in any event prior to deploying functionalities that are likely to have a critical impact on the risks identified pursuant to this Article. This risk assessment shall be specific to their services and proportionate to the systemic risks, taking into consideration their severity and probability, and shall include the following systemic risks:

(a) the dissemination of illegal content through their services;

(b) any actual or foreseeable negative effects for the exercise of fundamental rights, in particular the fundamental rights to human dignity enshrined in Article 1 of the Charter, to respect for private and family life enshrined in Article 7 of the Charter, to the protection of personal data enshrined in Article 8 of the Charter, to freedom of expression and information, including the freedom and pluralism of the media, enshrined in Article 11 of the Charter, to non-discrimination enshrined in Article 21 of the Charter, to respect for the rights of the child enshrined in Article 24 of the Charter and to a high-level of consumer protection enshrined in Article 38 of the Charter;

(c) any actual or foreseeable negative effects on civic discourse and electoral processes, and public security;

(d) any actual or foreseeable negative effects in relation to gender-based violence, the protection of public health and minors and serious negative consequences to the person's physical and mental well-being.

2. When conducting risk assessments, providers of very large online platforms and of very large online search engines shall take into account, in particular, whether and how the following factors influence any of the systemic risks referred to in paragraph 1:

(a) the design of their recommender systems and any other relevant algorithmic system;

(b) their content moderation systems;

(c) the applicable terms and conditions and their enforcement;

(d) systems for selecting and presenting advertisements;

(e) data related practices of the provider.

The assessments shall also analyse whether and how the risks pursuant to paragraph 1 are influenced by intentional manipulation of their service, including by inauthentic use or automated exploitation of the service, as well as the amplification and potentially rapid and wide dissemination of illegal content and of information that is incompatible with their terms and conditions.

The assessment shall take into account specific regional or linguistic aspects, including when specific to a Member State.

3. Providers of very large online platforms and of very large online search engines shall preserve the supporting documents of the risk assessments for at least three years after the performance of risk assessments, and shall, upon request, communicate them to the Commission and to the Digital Services Coordinator of establishment.